Chapter 3

Of Warps and Ripples

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hrough special relativity Einstein resolved the conflict between the "age-old intuition" about motion and the constancy of the speed of light. In short, the solution is that our intuition is wrong—it is informed by motion that typically is extremely slow compared to the speed of light, and such low speeds obscure the true character of space and time. Special relativity reveals their nature and shows them to differ radically from previous conceptions. Tinkering with our understanding of the foundations of space and time, though, was no small undertaking. Einstein soon realized that of the numerous reverberations following from the revelations of special relativity, one was especially profound: The dictum that nothing can outrun light proves to be incompatible with Newton's revered universal theory of gravity, proposed in the latter half of the seventeenth century. And so, while resolving one conflict, special relativity gave rise to another. After a decade of intense, sometimes tormented study, Einstein resolved the dilemma with his general theory of relativity. In this theory, Einstein once again revolutionized our understanding of space and time by showing that they warp and distort to communicate the force of gravity.

Newton's View of Gravity

Isaac Newton, born in 1642 in Lincolnshire, England, changed the face of scientific research by bringing the full force of mathematics to the service of physical inquiry. Newton's was such a monumental intellect that, for example, when he found that the mathematics required for some of his investigations did not exist, he invented it. Nearly three centuries would pass before the world would host a comparable scientific genius. Of Newton's numerous profound insights into the workings of the universe, the one that primarily concerns us here is his universal theory of gravity.

The force of gravity pervades everyday life. It keeps us and all of the objects around us fixed to the earth's surface; it keeps the air we breathe from escaping to outer space; it keeps the moon in orbit around the earth and it keeps the earth bound in orbit around the sun. Gravity